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Exp No.: 4 Querying Ledger Data with Hyperledger Fabric

Aim

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

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Procedure

Step 1: Navigate to the Fabric Test Network

Open the terminal and navigate to the Hyperledger Fabric test network.

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

Start the Fabric network and create the application channel.

```
./network.sh up createChannel -ca
```

Step 3: Deploy the Asset Transfer Chaincode

Deploy the JavaScript Asset Transfer chaincode.

```
./network.sh deployCC \
  -ccl javascript \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 4: Initialize the Ledger

Load the ledger with sample assets.

```
peer chaincode invoke ... -c '{"function":"InitLedger","Args":[]}'
```

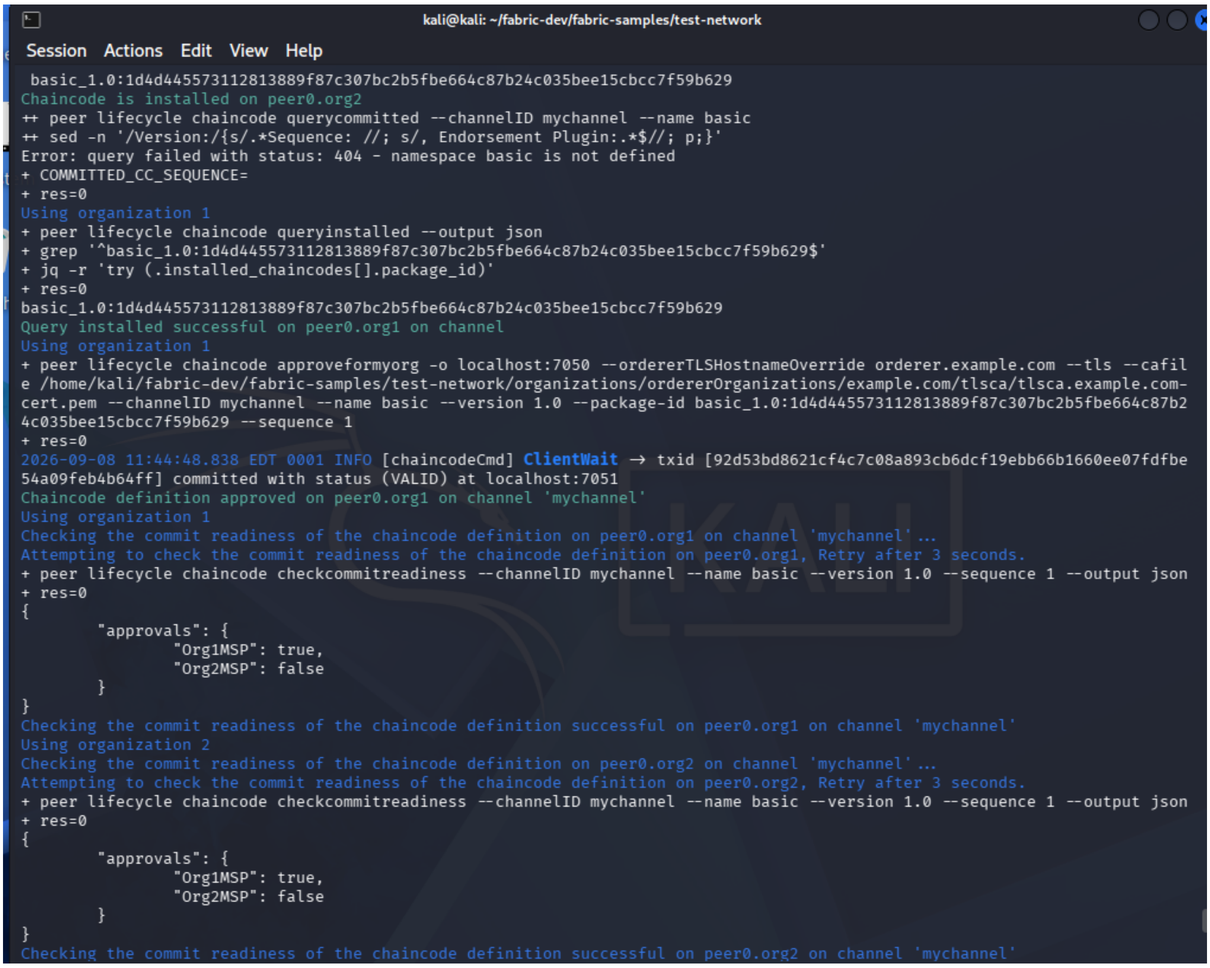

Figure 1: Chaincode installation, approval, and commit readiness check across both organizations

Step 5: Query All Assets

Retrieve all asset records stored in the blockchain ledger.

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

Step 6: Query a Specific Asset

Retrieve the details of asset1.

```
peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset","asset1"]}'
```

Step 7: Create a New Asset

Create a new asset (asset11) on the ledger.

```
peer chaincode invoke ... -c
'{"function":"CreateAsset","Args":["asset11","purple","12","Arun","900"]}'
```

Step 8: Query the Newly Created Asset

Read back asset11 to verify it was created.

```
peer chaincode query -C mychannel -n basic -c
'{"Args":["ReadAsset","asset11"]}'
```

Step 9: Stop the Fabric Network

Shut down all containers, volumes, and chaincode images.

```
./network.sh down
```

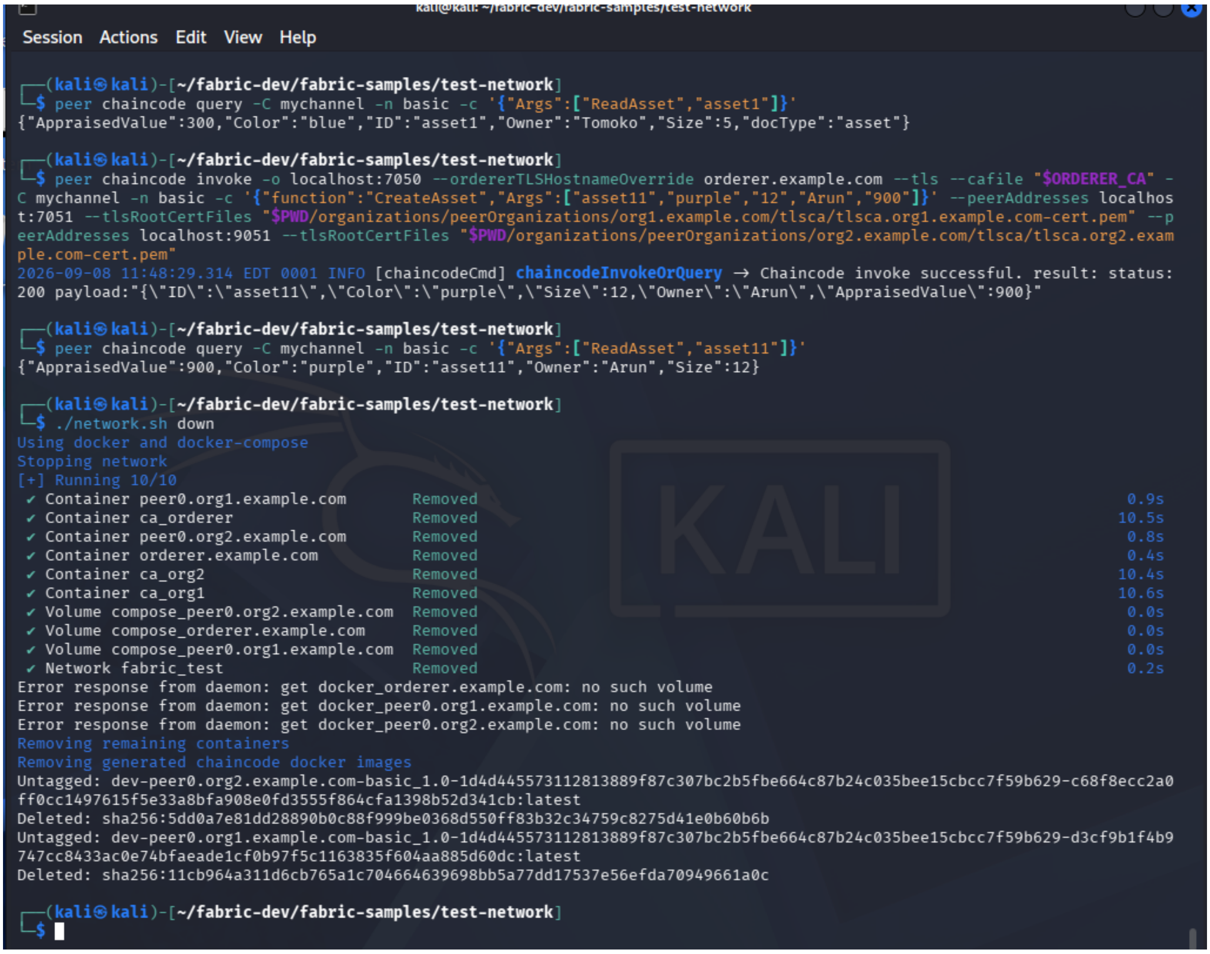

Figure 2: Querying asset1, creating asset11, verifying it, and stopping the network

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Flow of Ledger Query Process

Start Fabric Network
▼
Deploy Chaincode
▼
Initialize Ledger
▼
Query All Assets
▼
Query Specific Asset
▼
Create New Asset
▼
Query New Asset
▼
Verify Updated Ledger
▼
Stop Fabric Network

Result

The ledger data stored in the Hyperledger Fabric blockchain network was successfully queried using JavaScript chaincode. Asset records were retrieved from the world state database, specific asset details were verified, and newly created assets were successfully queried, demonstrating efficient data retrieval and verification in a permissioned blockchain environment.